

HUJI Applicative Research Report — Exact & Social Sciences — Week 38, 2026

1 paper(s) · Exact & Social Sciences · Weekly · Week 38, 2026 · Generated September 14, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This week's spotlight unveils a groundbreaking advancement in materials science that redefines how electron spins can be manipulated. Researchers at Hebrew University have uncovered the theoretical foundation for using chiral molecules as precise spin filters, fundamentally removing the reliance on bulky and energy-intensive external magnetic fields. This innovation promises to unlock a new generation of compact, energy-efficient spintronic devices and highly sensitive biosensors. The implications are far-reaching, from next-generation computing to portable medical diagnostics and clean energy applications.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

• Yossi Paltiel — Materials, Quantum, Diagnostics, Clean Energy

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Breaking reciprocity of the electron wave function in chiral materials with nonlocal interactions - the chiral-induced spin selectivity case. 20/50

"Molecular chirality enables compact spintronic devices and ultrasensitive sensors, eliminating the need for heavy magnets."

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ABOUT THIS RESEARCH

This research explores the physics of Chiral-Induced Spin Selectivity (CISS), which explains how chiral molecules act as spin filters for electrons. By demonstrating that nonlocal interactions break wave function reciprocity, the study provides a theoretical foundation for manipulating electron spins using molecular structures rather than external magnetic fields.

COMMERCIAL ANGLE

This discovery enables the development of spintronic devices and highly efficient electrochemical sensors that leverage molecular chirality to perform logic operations or sensitive biomolecular detection without traditional heavy magnets.

COMMERCIALISATION METRICS

Novelty

9/10

This research provides a fundamental theoretical breakthrough in the physics of chiral-induced spin selectivity (CISS), offering a mechanism to break time-reversal symmetry that could enable the development of high-efficiency, spin-polarized electronic and quantum computing devices at room temperature.

Commercial Potential

3/10

This is a fundamental physics paper addressing the theoretical mechanisms of spin selectivity, which is highly speculative and lacks a direct path to a defined technological product or industrial application. While Chiral-Induced Spin Selectivity (CISS) has long-term potential in spintronics, this specific work focuses on wave function reciprocity rather than a tangible device architecture.

Market Size

3/10

This is fundamental condensed matter physics research focused on quantum wave function mechanics, which lacks a direct or immediate path to a tangible commercial market. While chiral-induced spin selectivity (CISS) has potential applications in spintronics, the distance between this theoretical breakthrough and a scalable, addressable market remains vast.

Tech Readiness

2/10

This work is at the fundamental physics and theoretical modeling stage, focusing on the quantum mechanical behavior of electron wave functions in chiral materials. It addresses basic principles of spin selectivity rather than providing a prototype, scalable process, or industrial application.

IP Strength

3/10

This paper explores fundamental quantum mechanical properties and theoretical physics regarding electron wave functions, which generally constitute scientific discovery rather than commercially protectable intellectual property. The abstract nature of this work lacks the specific engineering embodiments or industrial applications necessary to establish a strong defensible moat in a deep-tech investment context.

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